AI UPCHAR

Complete Product Workflow & Functional Specification

Demo V2 • Patient & Operations First • Doctor Workspace UI Deferred

1. Product Objective

AI UPCHAR is built around a continuous patient journey. An appointment starts a guided AI intake; patient answers and uploaded documents build structured context; an appointment-specific AI brief is prepared for the doctor; the doctor remains responsible for final diagnosis, treatment and clinical decisions.

2. End-to-End Master Flow

Patient → Proceed with Appointment → New/Existing Patient → Appointment Details → Adaptive AI Conversation → Document Upload & Intelligence → Updated Patient Context → AI Pre-Visit Clinical Brief → Doctor → Consultation → Doctor Decision → Prescription / Tests / Scans / Referral → Billing / Inventory → Follow-up → Next Appointment → Previous Context reused.

3. Main Navigation

Navigation	Sections
Dashboard	Overview
Appointments	All; New; Follow-up; Today's Schedule; Calendar
Patients	All; New; Follow-up Due; Patient 360
Reports & Documents	All; Lab; Imaging; Prescriptions / Medical Documents
Inventory	Overview; Medicines; Tests; Scans; Stock; Stock Transactions
Billing	Revenue; Appointment Fees; Medicines; Tests & Scans; Patient-wise Transactions
Analytics	Operational and revenue analytics
Settings	System administration

4. Appointment Initiation

1. Patient clicks Proceed / Book Appointment.
2. AI conversation opens immediately.
3. AI identifies New or Existing patient.
4. AI collects appointment details and clinically relevant context one question at a time.
5. Every answer updates context and influences the next question.
6. Patient may upload relevant prescriptions, reports and scans during the conversation.
7. AI processes documents and may ask additional relevant questions.
8. Patient confirms collected information.
9. Appointment and its complete intake record are saved.

5. New Patient Flow

- New patient identification
- Mobile number
- Full name
- Age / DOB

- Gender
- Location
- Appointment date
- Preferred time
- Main complaint / reason
- Duration / onset
- Complaint-specific symptoms
- Relevant history
- Previous doctor consultation
- Previous prescription upload
- Previous reports / scans upload
- Other dynamically selected questions

6. Existing / Follow-up Flow

Known patient information is loaded first. The AI should not repeatedly ask information already known unless confirmation or change is relevant.

- Load previous visits, treatments, prescriptions, reports, tests and follow-up information.
- Ask why the patient is returning.
- Identify what changed since the previous visit.
- Check completion/results of previously advised tests.
- Request new documents when relevant.
- Ask only missing or changed information.
- Create a new appointment-specific transcript without overwriting previous history.

7. Adaptive AI Conversation Engine

The AI conversation is NOT a fixed questionnaire. It is a dynamic clinical-intake interview.

Patient Answer → Understand → Update Context → Identify Clinical Pathway → Safety/Red-Flag Check → Identify Missing High-Value Information → Rank Candidate Questions → Ask Best Next Question → Repeat.

- Complaint-specific questioning
- One question at a time
- Skip irrelevant questions
- Branch when important findings appear
- Use previous patient context
- Use uploaded-document findings
- Prioritize safety and information value
- Avoid generic unrelated questions

8. Example: Complaint-Specific Questioning

For: “Mera pet kharab hai, loose motion ho raha hai, 20 din se.” The engine should select relevant questions such as:

- Approximate stool frequency per day
- Watery/loose/other stool character
- Blood, black stool or mucus
- Fever/chills
- Abdominal pain/cramping and location

- Nausea/vomiting
- Possible dehydration symptoms
- Unintended weight loss
- Recent travel, outside food, unsafe water or food exposure
- Recent antibiotics or new medicines
- Relevant previous medical/GI history

This is an example of the adaptive principle, not a universal fixed script. The next question must depend on the patient's answers and context.

9. Clinical Knowledge / Evidence Layer

- Map the complaint to a relevant clinical pathway.
- Use curated medical knowledge and appropriate evidence retrieval.
- Generate candidate questions and considerations.
- Rank questions by relevance, safety importance and information value.
- Track provenance for important evidence-derived information.

10. Safety & Escalation Engine

Safety assessment runs throughout the conversation and can interrupt normal questioning.

- Routine
- Needs clinician review
- Prompt clinical attention
- Potential emergency / escalation flag

The AI surfaces safety concerns and urgency considerations; it does not autonomously make final admission, surgery, diagnosis or treatment decisions.

11. Document Upload & Intelligence

- Prescription photos/PDFs
- Lab reports
- X-ray / MRI / CT / Ultrasound reports
- Discharge summaries
- Referral letters
- Other medical documents

Pipeline: Upload → Classify → Extract → Structure → Link to Patient + Appointment → Update Context → Influence Next Question / Pre-Visit Brief.

12. Appointment-Specific Chat History

Every appointment gets its own persistent transcript. Historical transcripts are never overwritten.

- Appointment #1 → separate AI chat
- Appointment #2 → separate AI chat
- Appointment #3 → separate AI chat
- Each transcript links to patient + appointment.
- Patient profile shows all appointment conversations.
- Doctor-facing patient profile can view them.

- AI can use summarized prior conversations as future context.

13. Longitudinal Patient Context

- Identity
- Previous appointment dates/reasons
- Previous chat summaries
- Doctor notes where applicable
- Previous prescriptions/medicines
- Current treatment
- Tests/scans advised
- Reports and extracted findings
- Follow-up dates
- Relevant billing history

Raw transcripts remain available; structured summaries are used for efficient future context.

14. AI Pre-Visit Clinical Brief

- Patient information
- Appointment information
- Chief complaint
- History of present complaint
- Symptoms
- Relevant medical history
- Previous treatment
- Current medicines
- Previous consultations
- Uploaded documents
- AI-extracted document findings
- Changes since previous visit
- Safety/red-flag findings
- Missing information
- Clinical considerations / differential considerations
- Potential investigations to consider
- Urgency considerations
- Evidence/source context
- Doctor review and decision

15. AI Clinical Decision Support Boundary

AI may organize information, identify patterns, surface clinical considerations, highlight safety concerns and suggest questions or areas for clinician consideration. Final diagnosis, treatment, prescription, investigation orders, admission/disposition and surgery decisions remain with the doctor. AI suggestions must be visibly distinct from doctor-approved decisions.

16. Doctor Workspace — Future Phase

Current Doctor Workspace UI is removed/deferred. Underlying workflow logic and data relationships are preserved.

Future flow: Prepared Patient Context → AI Brief → History/Reports → Doctor Consultation → Assessment → Doctor Decision → Prescription / Tests / Scans / Referral → Review → Approve/Sign → Dispatch → Follow-up.

17. Prescription Letter

10. AI prepares a draft/context for the eventual clinician workflow.
11. Doctor reviews and edits.
12. Doctor makes final clinical decision.
13. Doctor approves/signs.
14. System creates final prescription letter.
15. Doctor can print it for the patient.
16. Approved prescription is stored under appointment and patient profile.
17. Patient can view the approved prescription in the patient portal.

AI draft ≠ final prescription. Only the doctor-approved/signed version is the official prescription.

18. Patient Portal

- Appointments
- Appointment chat history where appropriate
- Approved prescriptions
- Reports/documents
- Current medicines/treatment as applicable
- Tests/scans
- Follow-up information
- Billing/transaction history as applicable

19. Inventory

Inventory is a standalone operational module for catalogue, pricing and stock.

- Medicine Master
- Test Master
- Scan/Imaging Master
- Stock
- Stock Transactions

Medicine: Add / Edit / Delete / Purchase Price / Sale Price per Unit / Current Stock / Minimum Stock / Stock In / Stock Out / Returns / Adjustments / Low Stock / Out of Stock.

Tests and scans: Add / Edit / Delete / Price / Availability or operational status. Do not model MRI/test services as physical tablet-like stock.

20. Seed Medicine Catalogue

Medicine	Unit	Demo Sale Price
Paracetamol 500 mg	Tablet	₹2
Azithromycin 500 mg	Tablet	₹12
Amoxicillin 500 mg	Capsule	₹8
Pantoprazole 40 mg	Tablet	₹5
Cetirizine 10 mg	Tablet	₹3
Metformin 500 mg	Tablet	₹4
Amlodipine 5 mg	Tablet	₹3
Atorvastatin 10 mg	Tablet	₹6
Omeprazole 20 mg	Capsule	₹4
Ibuprofen 400 mg	Tablet	₹5
ORS	Sachet	₹20

Ondansetron 4 mg	Tablet	₹7
Diclofenac Gel	Tube	₹95
Cough Syrup	Bottle	₹110
Multivitamin	Tablet	₹6

Demo seed values only; prices are editable.

21. Seed Test & Scan Catalogue

Service	Category	Demo Price
CBC	Test	₹350
Blood Glucose	Test	₹100
HbA1c	Test	₹450
Lipid Profile	Test	₹650
Liver Function Test	Test	₹700
Kidney Function Test	Test	₹650
Thyroid Profile	Test	₹550
Urine Routine	Test	₹180
Vitamin D	Test	₹1,200
Vitamin B12	Test	₹800
X-Ray Chest	Scan	₹500
X-Ray Knee	Scan	₹500
Ultrasound Abdomen	Scan	₹1,000
Ultrasound Pelvis	Scan	₹1,000
CT Brain	Scan	₹3,500
CT Abdomen	Scan	₹5,000
MRI Brain	Scan	₹6,000
MRI Spine	Scan	₹7,000
ECG	Procedure	₹300
2D Echo	Scan	₹2,500

22. Stock Logic

- Stock In: Current Stock + received quantity
- Sale/Dispensing: Current Stock − dispensed/sold quantity
- Return: Current Stock + returned quantity
- Adjustment: controlled correction with a stock transaction record

Example: 100 units + 500 received = 600; 10 sold = 590.

23. Billing ↔ Patient ↔ Inventory

- Appointment fee creates an appointment billing transaction.
- Actual medicine dispensing/sale creates a billing transaction and reduces medicine stock.
- Prescription alone does not reduce inventory; actual dispensing/sale does.
- Tests/scans create patient-linked service orders and billing when applicable.
- Patient profile shows relevant billing history.
- Billing totals reconcile with underlying transactions.

24. Patient 360 Structure

- Overview
- Timeline / Patient History
- Appointments
- Appointment Chat History

- Reports & Documents
- AI Summary
- Medicines
- Current Treatment
- Prescriptions
- Tests / Scans
- Follow-up / Next Visit
- Billing / Transactions

25. Key Data Relationships

Patient → many Appointments. Appointment → appointment-specific AI conversation + documents + AI brief + clinical outputs + billing links. Patient → longitudinal summaries and all historical appointments. Prescription → appointment + patient. Medicine sale → patient/appointment where applicable + stock-out + billing transaction.

26. Data Consistency Rules

- Dashboard counts must match actual records.
- Patient reports must remain consistent across screens.
- Each chat belongs to exactly its appointment.
- Follow-up date must match across appointment and patient views.
- Billing totals must equal transaction sums.
- Medicine sales must reconcile with stock movements.
- AI-generated information keeps source/provenance where applicable.
- Doctor-approved information is distinguishable from AI suggestions.

27. Removed / Deferred

- Standalone AI Inbox — removed.
- Standalone Consultation navigation — removed; consultation belongs inside Appointment.
- Standalone Treatment & Orders navigation — removed; treatment belongs inside Patient/Appointment workflow.
- Old standalone AI Tool / random AI widget — removed.
- Current Doctor Workspace UI — deferred; underlying logic/data model preserved.

28. Implementation Order

18. Finalize data model and relationships.
19. Build one coherent demo dataset.
20. Rebuild Patient 360.
21. Rebuild Appointment creation/scheduling.
22. Build AI Conversation and adaptive question engine.
23. Build appointment-specific chat history.
24. Build document upload and intelligence.
25. Build AI Pre-Visit Brief.
26. Build Inventory and stock transactions.
27. Connect Billing with Patient, Appointment and Inventory.
28. Validate cross-screen consistency.
29. Design Doctor Workspace UI only after the patient/operations side is complete.

29. Final Architecture Principle

Navigation = major destinations.
Appointment = a complete visit episode including its AI conversation and preparation.
Patient Profile = the complete longitudinal record.
AI = an embedded intelligence layer that understands context and selects relevant next questions.
Documents = inputs to context building and adaptive questioning.
Prescription = doctor-approved official output.
Inventory = master catalogue + operational stock management.
Doctor Workspace = future UI layer using preserved logic and data.